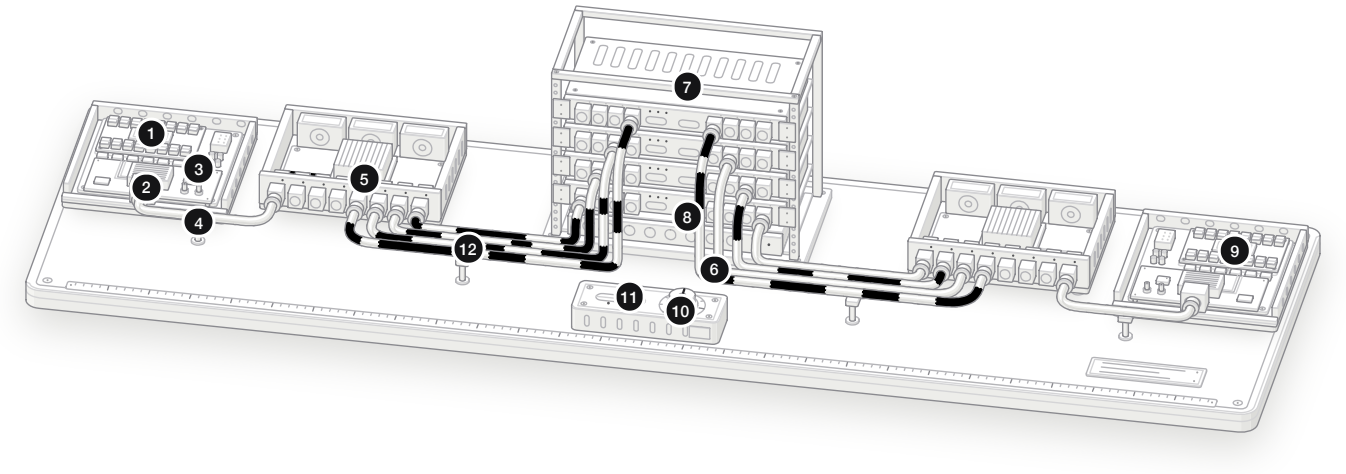
The AI network stack

AN-002
HOW IT WORKS
HOST, FABRIC, LINK, AND WHO WRITES THE RULES FOR EACH



What does each spec do to a message crossing an AI network?

OCP NIC Core Features sets the host side of the card: queues, interrupts, timestamps, pacing. Ultra Ethernet 1.1 sets the scale-out transport (SES cuts packets, PDS makes them reliable, CMS controls congestion and sprays them) and the link features LLR and CBFC. OCP ESUN takes those link features into the scale-up domain behind a 4-byte header. Here a 64 MiB message crosses four spines while elephant flows from other jobs fill two of them.

6×
FASTER SPRAYED THAN ONE HASHED PATH, 3 ELEPHANTS

4096 B
PAYLOAD PER PACKET, UE DEFAULT

4 B
ESUN HEADER, REPLACING 20–40 B OF IP

800 Gb/s
NIC LINK, OVER 4 × 400G SPINE PATHS

01 **Accelerator memory**
16 HBM stacks hold the 64 MiB message; each cell is 1/16 of it.

02 **UE NIC**
OCP queues and DMA to the host; UE SES cuts 4 KiB packets, PDS numbers them, CMS picks each path.

03 **PTP clock and PPS jacks**
OCP: ≤10 ns hardware timestamps, PPS in and out; UE's NSCC times every round trip with them.

04 **800G NIC links**
Copper DAC; UE's LLR resends a corrupted frame from the link's own replay buffer.

05 **Leaf switches**
Plain ECMP on each packet's entropy value; ECN marked on dequeue; may trim, not drop, when full.

06 **Spine links, 4 × 400G**
A flow-hashed transfer is stuck on one; sprayed packets use every free one.

07 **Spine switches**
Four spines; elephant flows from other jobs fill spines 2 and 4 here.

08 **Rack and PDU**
Posts and rails hold the spines; the PDU feeds them power.

09 **Receiving memory**
UE's RUD writes each packet straight to its offset, so arrival order doesn't matter.

10 **Elephant knob**
0–6 elephant flows already pinned to spines by ECMP.

11 **Mode switch**
One flow-hashed path, or sprayed (UE RUD).

12 **Cable ladders**
Carry the long runs at port height.